

Seven-segment

A seven-segment display turns a four-bit nibble into a glyph, zero through nine and A through F

Segments, patterns, polarity

- Starter nibble A in common cathode drives one-one-one-zero-one-one-one
- Digit eight lights all seven; digit one lights only b and c on the right
- Digit zero turns g off in the middle
- Flip to common anode and the same glyph needs the bitwise complement on the drive bus
- The decoder is a lookup table, ROM or case statement in RTL, not seven unrelated wires

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Seven-segment decoder

Map a hex nibble to segments **a-g**. See the glyph, the bit vector, and common-cathode vs anode polarity.

Starter example: nibble 0xA → glyph A, common-cathode abcdefg = 1110111 (all but d).

Load starter example

Challenges (1/22)

Quiz: segments. A classic digit has how many segments? Answer: 7

Your answer

Show hintCheckNextIdle

quiz-countquiz-nibblequiz-ccquiz-castarterstarter-d-offshow-8show-1show-0show-fanode

anode-a pick-5 explain quiz-order quiz-8 b-only invert-eq glyph-b quiz-rom g-on-8 full-path

Core ideas

Nibble

4-bit code 0...F selects one of 16 glyphs.

a-g

Seven segments; decoder is a ROM / SOP cover.

Polarity

Common-cathode: 1=ON. Anode: invert the bus.

Seven segment starter

learn_digital — seven segment

Workbook practice

- In the workbook track
- Write the common-cathode pattern for F
- For nibble A on common anode, give the inverted drive string
- Name one pitfall

Pitfalls to watch

- Do not scramble segment order, this lab uses abcdefg as the bit string
- Mixed-case hex letters are for readability, not arbitrary encoding
- And remember: the browser lab is literacy
- Real displays still need current limiting, multiplexing

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, draw one digit and write its abcdefg pattern
- When you are ready, take the short quiz, then continue to clock edge stepper